

William Chang

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EDUCATION

University of Waterloo

Bachelor's in Honours Mathematics, Combinatorics and Optimization

Waterloo, ON, Canada

September 2020 - May 2025

- Relevant Coursework: Algorithms & Data Structures, Object-Oriented Design & Programming, Databases

AWS Certifications

AWS Certified Cloud Practitioner

Online

November 2023

- Route 53, Lambda, S3, ELB, SNS, RDS, EC2, Cloud Concepts, Security, Billing & Pricing

TECHNICAL SKILLS

Languages: C++, Python, Java, Javascript, Typescript

Frameworks and Databases: Bootstrap, Django, Kafka, Next.js, React, Redis, MySQL, SQLite

Tools: Git, Github Actions, FramerMotion, Tailwind CSS, Vercel

WORK EXPERIENCE

Schedulancer — Next.js, GitHub Actions, Kafka, Redis, Typescript, Tailwind CSS

Waterloo, ON, Canada

Co-Founder

June 2023 - Present

- Built an online scheduling service targeting freelancers with pre-existing clientele bases at a lower commission rate relative to competitor websites using a **React** framework, **Next.js** and **Tailwind CSS**
- Created a payment gateway where service providers can send payment requests to clients by integrating **Stripe's** webhook endpoint to publish events into **Kafka** to monitor payments in real-time
- Developed a user authentication page with OAuth providers, a scheduling system with a calendar view, a user matching system, and a messaging service using **Auth0**, **Redis**, **Kafka**, and **Typescript**
- Seen user growth of over **1500%** since initial launch with around **50** payment transactions per month
- Utilized **object oriented design** to adhere to **system design** principles while also applying unit and integral tests
- Systematized the workload with the **Agile SDLC** method and collaborated with co-founder in **Github Actions** for continuous integration and deployment after performing routine functional tests

Lee's Orthokeratology Clinic — Python, MySQL

Vancouver, BC, Canada

Data Analyst Intern

June 2022 - August 2022

- Spearheaded a new patient scheduling proposal using the data gathered with **Python** and **MySQL** which led to an increase in patient care efficiency, allowing for more clients within a day
- Used DrChrono to keep track of patient information, clinical charting, scheduling, and payments

City of Burnaby

Burnaby, BC, Canada

Lifeguard/Swim Instructor

November 2018 - December 2022

- Organized 8-week lesson plans for varying skill levels from toddlers to adults according to given criteria
- Handled public relations problems and complaints in Mandarin and French

PROJECTS

Website Portfolio — FramerMotion, Next.js, React, Tailwind CSS, Typescript, Vercel

- Utilized **Next.js** to server render **React** applications with interactive UIs to create a website portfolio
- Incorporated **FramerMotion** and **Tailwind CSS** to front-end components while following UI/UX principles
- Optimized resolution sizes to be compatible for screens from mobile to desktop viewers
- Applied **Next.js** revalidate function to optimize page rendering via incremental static regeneration

Student Management System — Bootstrap, Django, Python, SQLite

- Built a student management system that creates, reads, updates, and deletes student information using **Django** as the back-end framework accompanied by **Bootstrap** and **Bootstrap** for the front-end
- Stored student data in a **SQLite** table where it can be viewed in DB browser, enabling ease of storage access

Pathfinding Visualizer — Javascript, React

- Created a **React** based pathfinding visualizer between two nodes using Dijkstra's algorithm where the user is able to create mazes and blockades to find the shortest path
- Designed UI buttons, grid, node, and path animations with **Javascript**, CSS, and HTML

Google Search Clone — React, RapidAPI, Tailwind CSS

- Designed and implemented **React** based Google search clone with the help of **Tailwind CSS**
- Used **RapidAPI** to mimic Google's search algorithm including links, titles, SERP, images, videos, and news